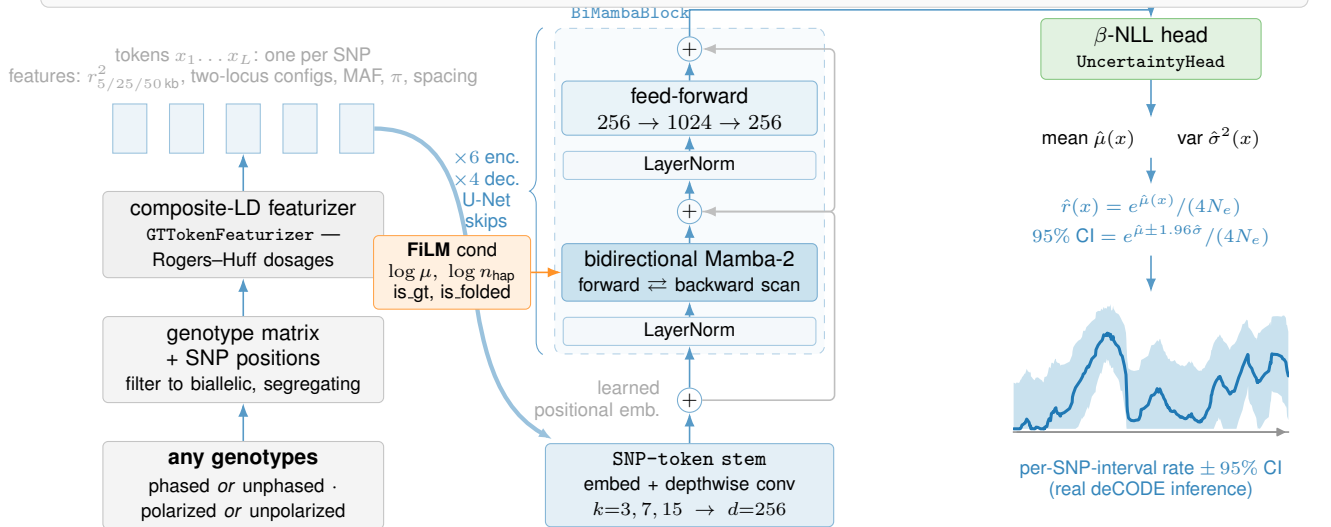


Trained once (amortized). msprime + stdpopsim prior — demographics (constant / bottleneck / sawtooth / island) $\times$ sample sizes $\times$ hotspot maps $\rightarrow$ domain-randomized corpus $\rightarrow$ **one frozen model**, applied to any dataset with no lookup table, no per-dataset tuning, no retraining



a any genotypes $\rightarrow$ tokens

b amortized bidirectional SSM (Mamba-2)

c calibrated map in one forward pass